

Chapter 5: The Data Link Layer

Our goals:

- ❑ principles behind data link layer services:
 - error detection, correction
 - sharing a broadcast channel: multiple access
 - link layer addressing
 - reliable data transfer, flow control: *done!*
- ❑ instantiation and implementation of various link layer technologies

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Data Link Layer (SSL) 5-1

Link layer, LANs

5.1 introduction, services

5.2 error detection, correction

5.3 multiple access protocols

5.4 LANs

- addressing, ARP
- Ethernet
- switches
- VLANs

5.5 link virtualization: MPLS

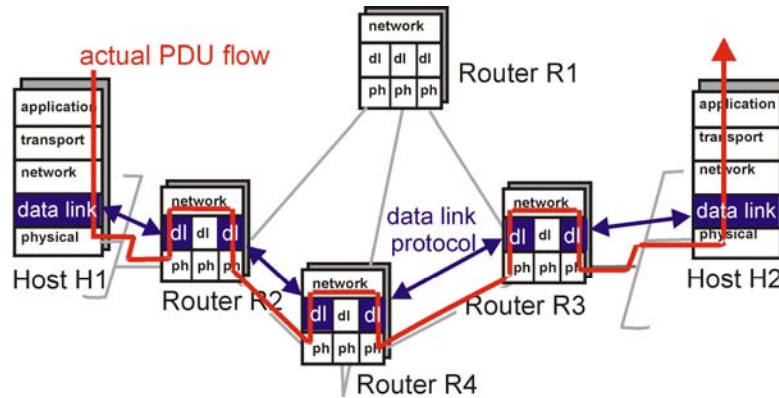
5.6 data center networks

5.7 a day in the life of a web request (play animation in .ppt slides on your own)

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Data Link Layer (SSL) 5-2

Link Layer: context



- A link connects two adjacent IP nodes (layer 3) along a path

- IP datagram transferred by different link protocols over different links which may provide different services

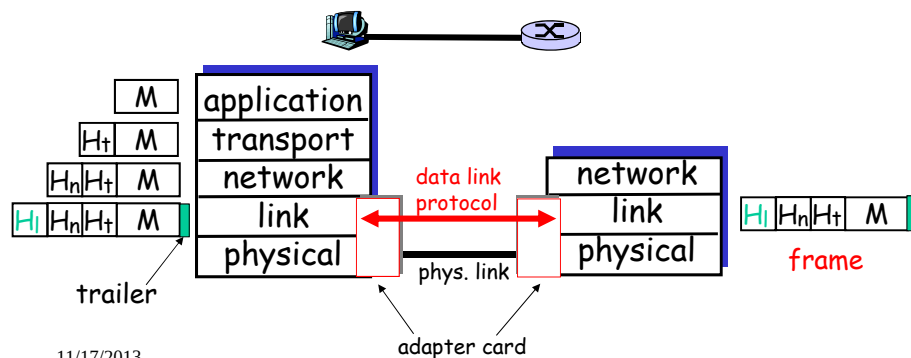
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Link Layer: context

- unit of data: *frame*, which encapsulates an IP datagram
- IP expects no service guarantee from links

- Link can be
 - wire
 - wireless
 - LAN (layer 2)
 - WAN (virtual link)



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Link Layer Services

- ❑ Framing
 - Encapsulate datagram with header and trailer
- ❑ Error Detection
 - errors caused by signal attenuation, noise.
 - receiver detects presence of errors
- ❑ Error Correction
 - receiver identifies *and corrects* bit error(s) without resorting to retransmission
- ❑ Link access
 - access protocol for shared channel access
 - "MAC" addresses used in frame headers to identify source, destination
 - different from IP addresses
 - why both MAC and IP addresses?

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Link Layer Services (more)

- ❑ Half-duplex and full-duplex
 - with half duplex, nodes at both ends of link can transmit, but *not* at same time
- ❑ Flow Control
 - pacing between sender and receiver(s)
- ❑ Reliable delivery between two physically connected devices
 - we learned how to do this already (chapter 3)
 - seldom used on low error-rate links (fiber, some twisted pair)
 - wireless links: high error rates
 - Q: why both link-level and end-end reliability?

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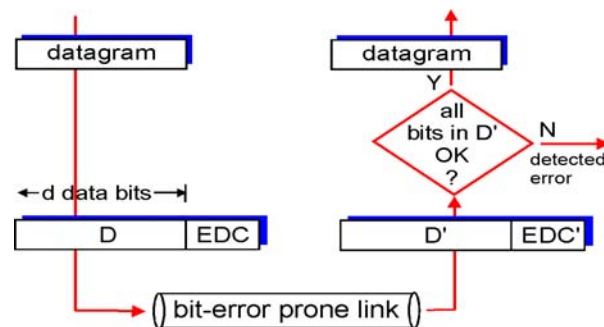
(play animation in .ppt slides on your own)

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Error Detection

EDC= Error Detection and Correction bits (redundant check bits)
D = Data protected by error checking, may include header fields



- Error detection not 100% reliable
 - protocol may miss some errors
 - more check bits yield better detection and correction

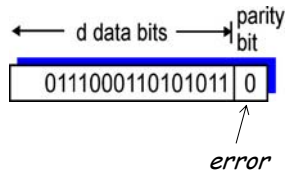
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Parity Checking

Single Bit Parity:

Detect single bit errors

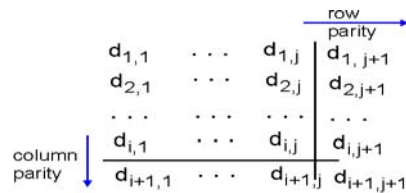


*Even parity:
total number of 1s
is odd means there
is a bit error or,
more precisely, an
odd number of bit
errors*

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Two Dimensional Bit Parity:

Detect and correct single bit errors



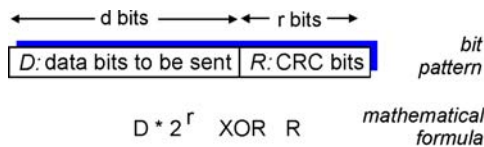
101011
111100
011101
001010
no errors

101011
101100
011101
001010
parity error
correctable
single bit error

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Cyclic Redundancy Check (CRC) - sender

- View data bits, D , as a binary number



- Choose $r+1$ bit pattern (generator), G

- Goal:** choose r CRC bits, R , such that $\langle D, R \rangle$ is exactly divisible by G using modulo 2 arithmetic

- Modulo 2 arithmetic**
 - there is no carry in addition, and no borrow in subtraction
 - addition and subtraction same as bitwise exclusive OR (XOR)

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Cyclic Redundancy Check (CRC) - receiver

□ Bit string $\langle D, R \rangle$ sent
is *exactly* divisible by
 G

□ Receiver knows G ,
performs division. If
non-zero remainder,
error detected!

❖ can detect all burst
errors less than $r+1$
bits;

❖ longer burst errors
are detectable with
probability $1-(0.5)^r$

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CRC Theory and Example

Want:

$$(D \cdot 2^r) \text{ XOR } R = nG$$

add R to both sides:

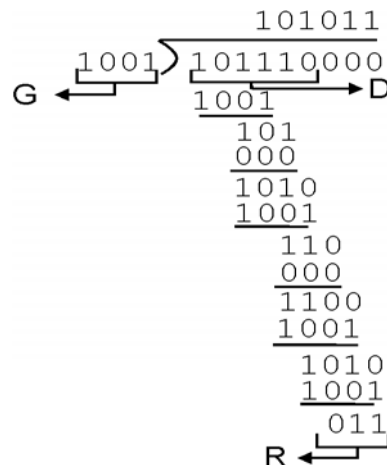
$$D \cdot 2^r = (nG) \text{ XOR } R$$

Equivalently:

the remainder from
dividing $D \cdot 2^r$ by G is
equal to R ;

the desired CRC bit
string is

$$R = \text{remainder} \left[\frac{D \cdot 2^r}{G} \right]$$



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Link layer, LANs

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protocols

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5.5 link virtualization:
MPLS

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(play animation in .ppt
slides on your own)

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Multiple Access Links and Protocols

Two types of "links":

□ point-to-point

- fiber optic link
- link between Ethernet switch and host

□ broadcast (shared wire or medium)

- old-fashioned Ethernet
- upstream channel of HFC (hybrid fiber cable)
- wireless (802.11 LAN and others), etc.



shared cable (e.g.,
old Ethernet)

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shared RF
(e.g., 802.11 WiFi)



shared RF
(satellite)



humans at a party
(shared air, acoustics)

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Multiple Access protocols

single shared broadcast channel

- two or more simultaneous transmissions by nodes may **interfere** with each other
 - **collision** if a node receives two or more signals at the same time
- Need a protocol to determine when nodes can transmit
 - no out-of-band channel for coordination

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Ideal/ Multiple Access Protocol

Broadcast channel of rate R bps

1. when one node wants to transmit, it can send at rate R .
2. when M nodes want to transmit, each can send at average rate R/M
3. fully decentralized:
 - no special node to coordinate transmissions
 - no synchronization of clocks, time slots
4. simple

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MA Protocols: a taxonomy

Three broad classes:

❑ Channel Partitioning

- divide channel into smaller "pieces" (time slots, frequency bands, codes)
- allocate a piece to each node for exclusive use

❑ Random Access

- shared channel, collisions allowed
- "recover" from collisions

❑ "Taking turns"

- nodes take turns
- a node with more to send can take a longer turn

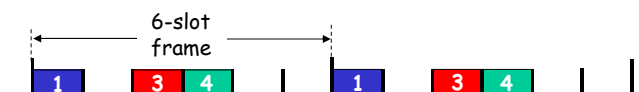
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Channel Partitioning protocols

TDMA: time division *multiple access**

- ❑ each station gets fixed length slot (length = pkt trans time) in each frame
 - requires time synchronization
- ❑ unused slots go idle



** there are multiple transmitters*

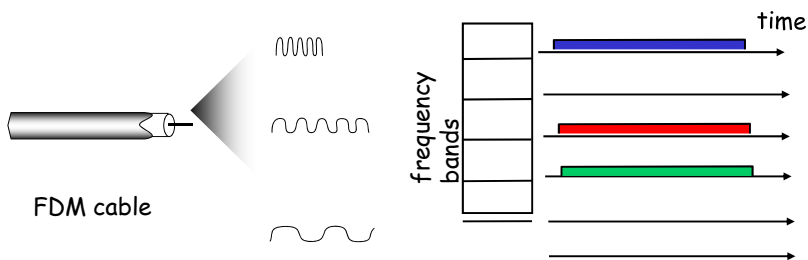
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Channel Partitioning protocols

FDMA: frequency division *multiple access*

- ❑ each station assigned a fixed frequency band
- ❑ unused transmission time in frequency bands go idle



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Random Access Protocols

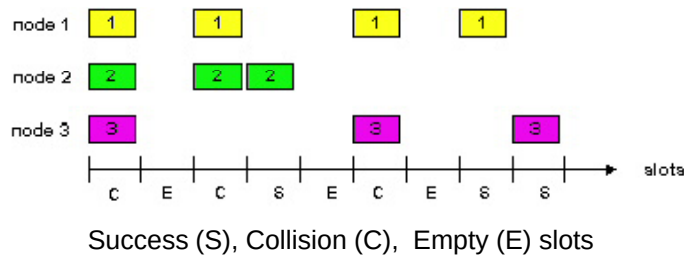
- ❑ When node has packet to send
 - transmit at full channel data rate
 - no *a priori* coordination among nodes
- ❑ two or more transmitting nodes → "collision"
- ❑ random access MA protocol specifies:
 - how to detect collision
 - how to recover from collision (e.g., via delayed retransmissions)
- ❑ examples:
 - slotted ALOHA
 - ALOHA
 - CSMA, CSMA/CD, CSMA/CA

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Slotted Aloha

- time is divided into equal size slots (pkt trans. times)
 - requires time synchronization
- node with new arriving pkt: transmit at beginning of next slot
- if collision: retransmit pkt in a future slot with probability p (or one of K slots at random), until successful.



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Slotted Aloha efficiency

Long-term fraction of time slots that are successful?

Suppose N nodes have packets to send

- each transmits in slot with probability p
- prob. successful transmission S is

by a particular node: $S = p(1-p)^{N-1}$

by any of N nodes:

$S = \text{Prob} [\text{one of } N \text{ nodes transmits}]$

$$= Np(1-p)^{N-1}$$

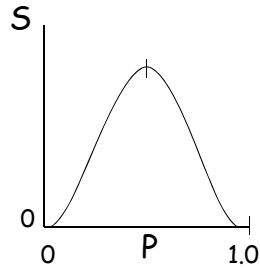
... choosing optimum p , let $N \rightarrow \text{infinity}$

$$= 1/e = .37 \text{ as } N \rightarrow \text{infinity}$$

Channel occupied by useful transmissions < 37% of time

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$$\frac{\partial S}{\partial P} = \frac{\partial}{\partial P} [NP(1-P)^{N-1}]$$

$$= NP \frac{\partial}{\partial P} (1-P)^{N-1} + (1-P)^{N-1} N$$

$$= -NP(N-1)(1-P)^{N-2} + N(1-P)^{N-1}$$

$$= N(1-P)^{N-2} \{-P(N-1) + 1 - P\}$$

$$= N(1-P)^{N-2} \{-NP + P + 1 - P\}$$

$$\frac{\partial S}{\partial P} = 0 \text{ when } P = \frac{1}{N} \text{ to maximize } S$$

My terminology : "Probability Division Multiplex"

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$$S_{\max} = NP(1-P)^{N-1} \Big|_{P=\frac{1}{N}}$$

$$= N \left(\frac{1}{N} \right) \left(1 - \frac{1}{N} \right)^{N-1}$$

$$= \left(1 - \frac{1}{N} \right)^{N-1} \xrightarrow{N \rightarrow \infty} e^{-1}$$

$$\frac{1}{e} \cong 0.368$$

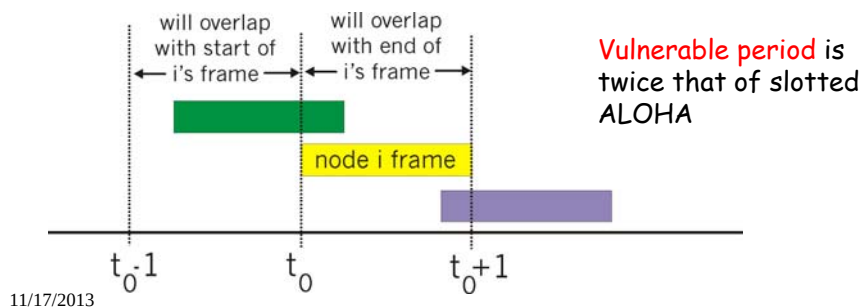
which is maximum throughput
(efficiency) of the slotted ALOHA
protocol

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Pure (unslotted) ALOHA

- ❑ unslotted Aloha: no time synchronization
- ❑ when frame arrives
 - send immediately (without waiting for beginning of slot)
- ❑ collision probability increases:
 - frame sent at t_0 can collide with another frame sent within $[t_0-1, t_0+1]$



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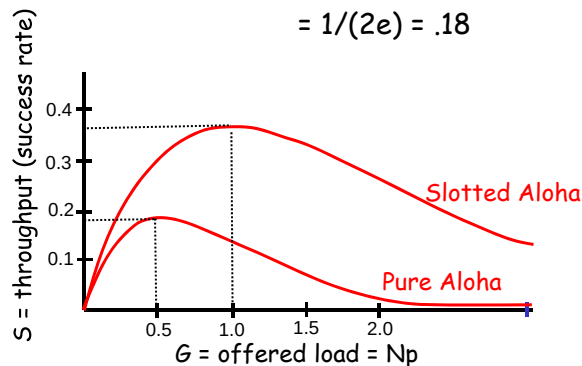
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Pure Aloha (cont.)

$P(\text{success by any of } N \text{ nodes})$

... choosing optimum p , let $N \rightarrow \text{infinity}$...

$$= 1/(2e) = .18$$



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CSMA: Carrier Sense Multiple Access

CSMA: listen before transmit:

- ❑ If channel sensed idle: transmit entire packet
- ❑ If channel sensed busy, **defer** transmission;
retry after some random interval
- ❑ human analogy: don't interrupt when someone else is speaking

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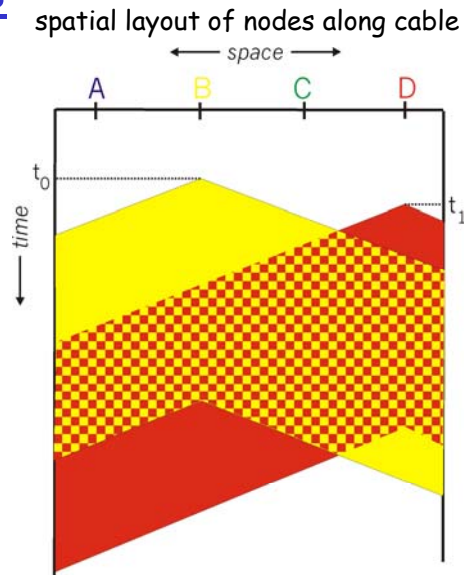
CSMA collisions

collisions can occur:

it takes time for two nodes to hear each other's transmission due to propagation delay

collision:

entire packet transmission time wasted



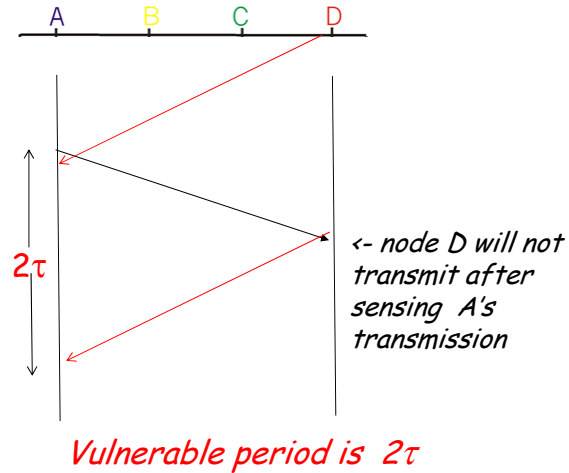
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Vulnerable period of a transmission

Let τ be the maximum one-way propagation delay between two nodes in a subnet

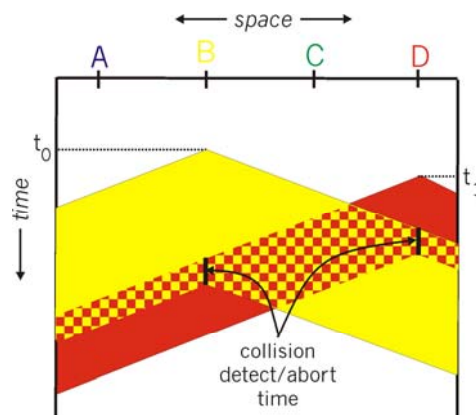
If sender A detects no collision after 2τ seconds, then it knows that its transmission will be successful



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CSMA/CD collision detection (& abort)



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CSMA/CD (Collision Detection)

Carrier sensing, deferral as in CSMA

- CD useful for channels where collisions are detectable within a *short* time
- *colliding transmissions aborted*, reducing channel wastage
- collision detection is
 - *easy* in *wired* LANs: measure signal strength, compare transmitted and received signals
 - *difficult* in *wireless* LANs: received signal overwhelmed by local transmission signal
- high channel utilization possible by sending very long packets (*relative to propagation delay*)

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CSMA/CD channel efficiency

Channel efficiency = $t_{\text{trans}} / (\text{contention period} + t_{\text{trans}})$

where t_{trans} is *average transmission time* of a frame

Let t_{prop} denote the maximum propagation delay between any two nodes. Then a good estimate of the average contention period is $2e t_{\text{prop}}$. (Why ?)

CSMA/CD channel efficiency = $t_{\text{trans}} / (2e t_{\text{prop}} + t_{\text{trans}})$

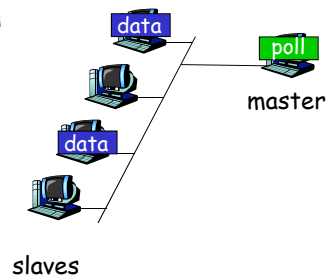
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"Taking Turns" MA protocols

Polling:

- ❑ master node "invites" slave nodes to transmit in turn
- ❑ concerns:
 - polling overhead
 - latency (for large N)
 - single point of failure (master)



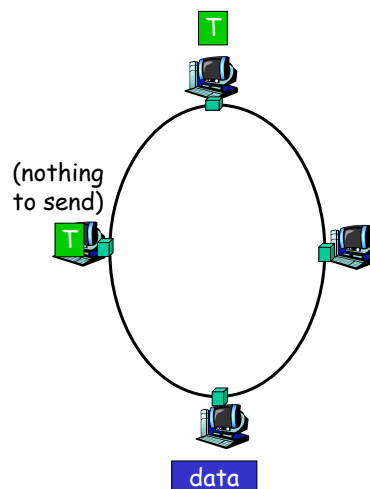
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"Taking Turns" MA protocols

Token passing:

- ❑ control **token** (short msg) passed from one node to next sequentially.
- ❑ Data removed from ring by its sender
 - => broadcast
- ❑ concerns:
 - ❖ latency (for large N)
 - ❖ single point of failure
 - active ring interface
 - token loss

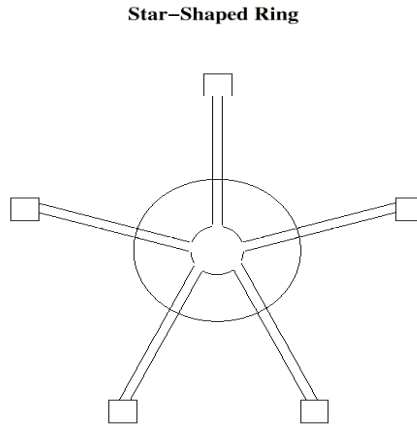


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Star-shaped Ring Topology

Example:
Token ring
(IEEE 802.5)
with wiring
closet



*Today's
Ethernet uses
a star topology*

Generalization: A hierarchical ring (with multiple wiring centers to reduce cable length).

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Summary of MA protocols

- ❑ *channel partitioning*, by time, frequency (or code)
 - Time Division, Frequency Division (also CDMA)
- ❑ *random access*
 - ALOHA, slotted ALOHA, CSMA, CSMA/CD
 - collision detection: easy in some technologies (wire), hard in others (wireless)
 - CSMA/CD used in Ethernet
 - CSMA/CA used in 802.11 (Chap. 6)
- ❑ *taking turns*
 - polling by a central site, e.g., Bluetooth
 - token passing - IBM Token Ring, FDDI

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(play animation in .ppt slides on your own)

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MAC and IP Addresses

□ 32-bit IP address:

- *network-layer address*
- *used to get datagram to destination IP subnet*

□ 48 bit MAC address (or LAN or *Ethernet* or link-layer address):

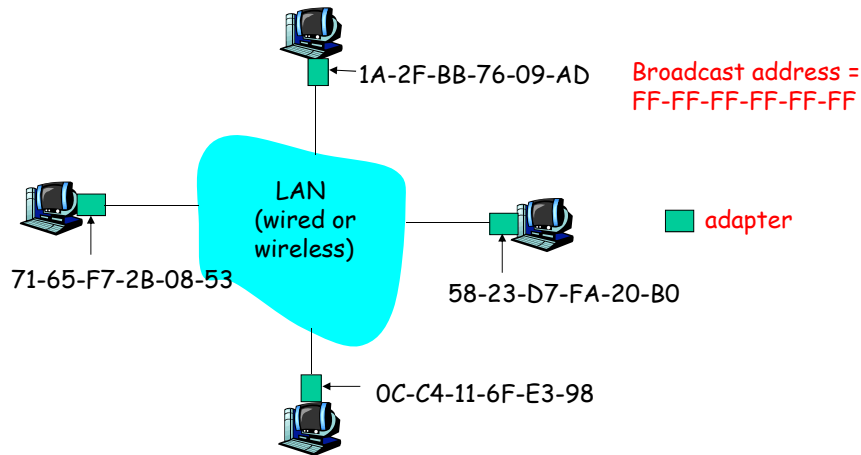
- e.g.: 1A-2F-BB-76-09-AD (*hexadecimal notation*)
- *burned in NIC ROM (also sometimes software settable)*
- *used to get frame from one interface to another interface in same subnet*

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MAC Addresses

Each adapter on LAN has unique MAC address



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MAC Address vs. IP address

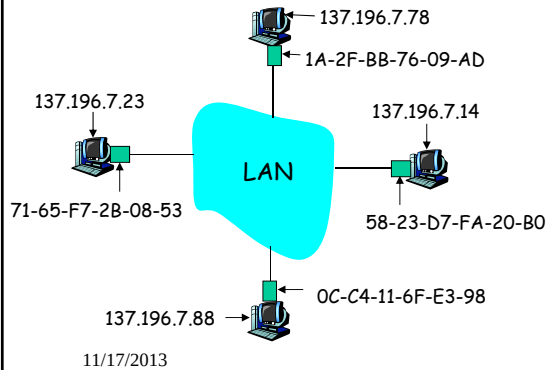
- ❑ MAC addresses are flat
 - MAC address allocation administered by IEEE
 - manufacturers buy blocks of MAC address space for a nominal fee
 - MAC addresses are portable
 - LAN card can be moved from one LAN to another, e.g., laptop
- ❑ IP's hierarchical address NOT portable
 - address depends on IP subnet to which node is attached
- ❑ analogy:
 - (a) MAC address: like Social Security Number
 - (b) IP address: like postal address

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ARP: Address Resolution Protocol

Question: how to determine MAC address of interface B knowing B's IP address?



- Each IP node (host, router) on LAN has **ARP** table
- ARP table: IP-MAC address mappings for some LAN nodes
 - < IP address; MAC address; TTL >
 - TTL (Time To Live): time after which address mapping will be forgotten (typically 20 min)

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ARP protocol: Same LAN

- A wants to send datagram to B, and B's MAC address *not* in A's ARP table.
- A **broadcasts** ARP query packet, containing B's IP address
 - Dest MAC address = FF-FF-FF-FF-FF-FF
 - all machines on LAN receive ARP query
- B receives ARP packet, replies to A with its (B's) MAC address
 - frame sent to A's MAC address (unicast)
- A caches IP-to-MAC address pair in its **ARP table**
 - **soft state**: information that times out (goes away) unless refreshed
- ARP is "plug-and-play":
 - nodes create their ARP tables without any work by net administrator

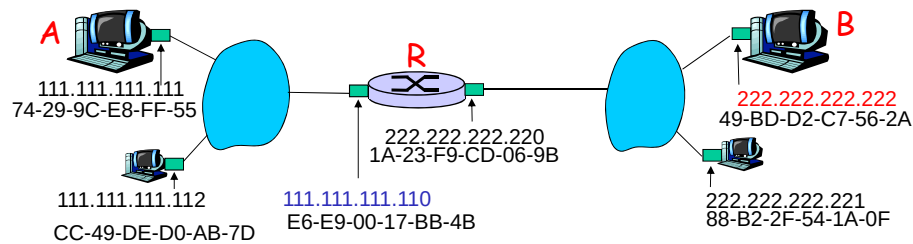
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Addressing: routing to another LAN

walkthrough: **A sends datagram to B via R.**

- focus on addressing - at both IP (datagram) and MAC layer (frame)
- A knows B's IP address
- A knows IP address of first-hop router, R
- A knows MAC address of first hop router's interface (how?)

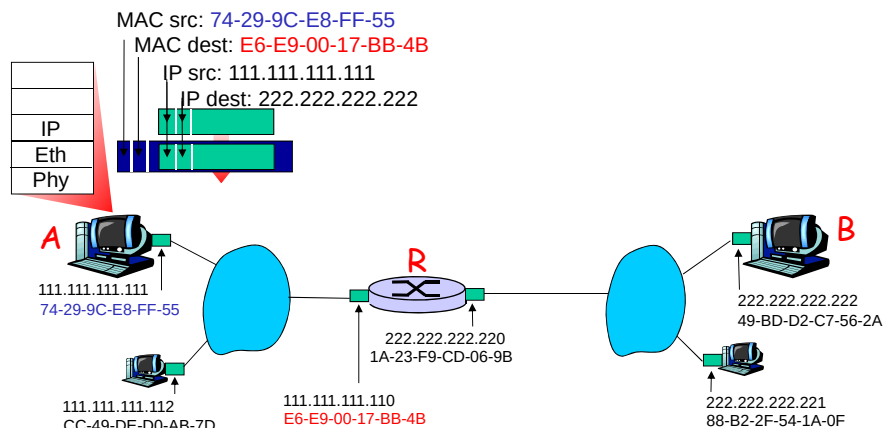


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Addressing: routing to another LAN

- ❖ A creates IP datagram with IP source A, destination B
- ❖ A creates link-layer frame with R's MAC address as dest, frame contains A-to-B IP datagram

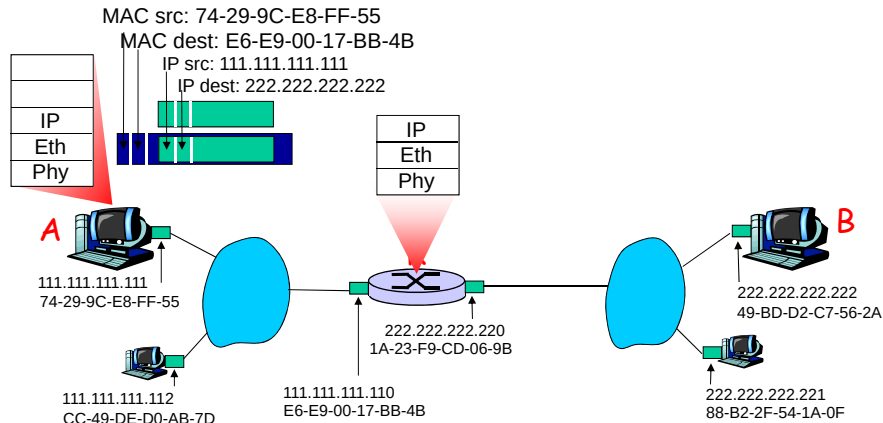


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Addressing: routing to another LAN

- ❖ frame sent from A to R
- ❖ frame received at R, datagram passed up to IP

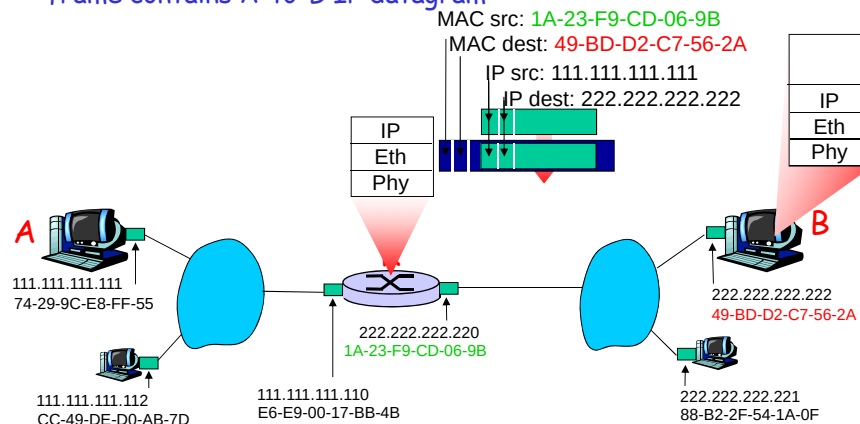


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Addressing: routing to another LAN

- ❖ R forwards datagram with IP source A, destination B
- ❖ R looks up B's MAC address
- ❖ R creates link-layer frame with B's MAC address as dest, frame contains A-to-B IP datagram

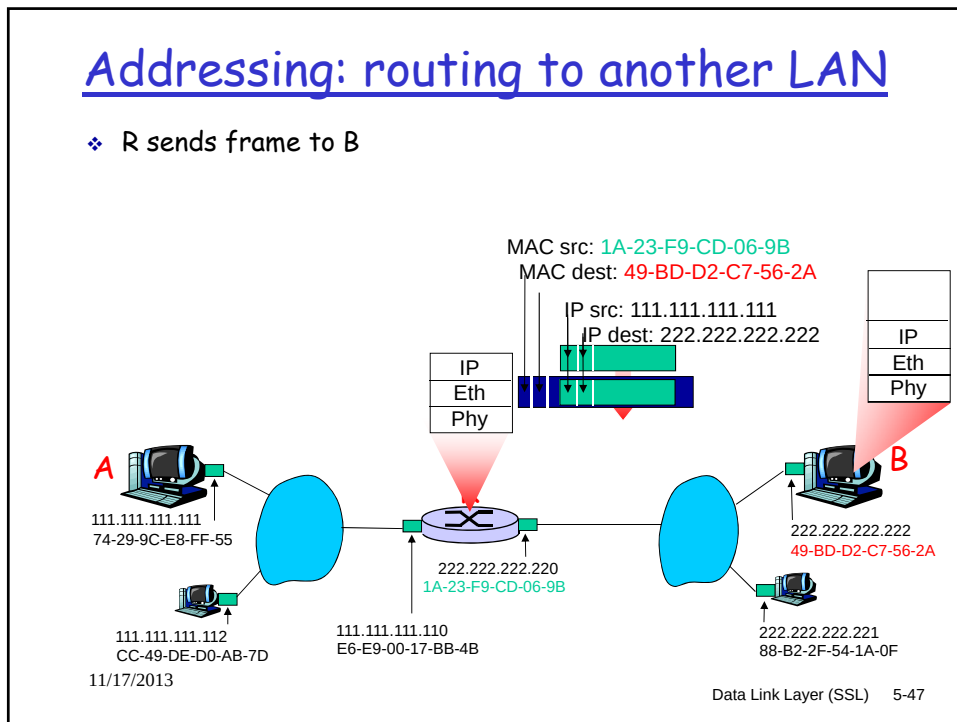


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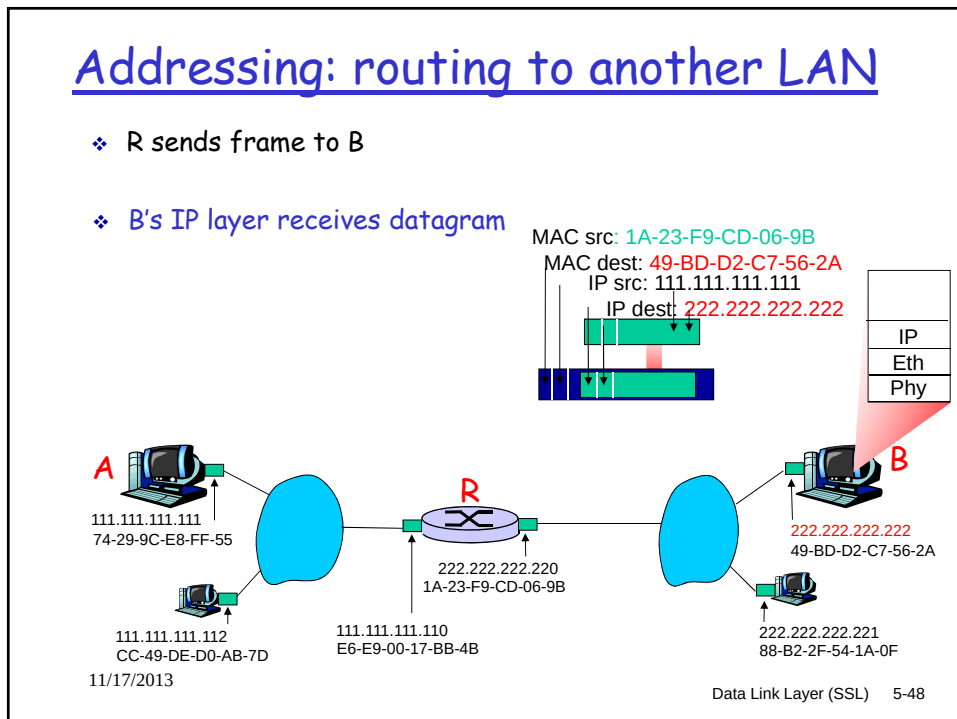
Addressing: routing to another LAN

- ❖ R sends frame to B



Addressing: routing to another LAN

- ❖ R sends frame to B
- ❖ B's IP layer receives datagram



Link layer, LANs

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(play animation in .ppt slides on your own)

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Ethernet

"dominant" wired LAN technology:

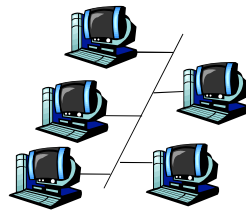
- ❑ cheap, \$20 for NIC
- ❑ first widely used LAN technology
- ❑ simpler, cheaper than competitors
 - token-ring (16 Mbps), FDDI (100 Mbps), and ATM (155 Mbps)
- ❑ kept up with speed race: 10 Mbps - 10 Gbps

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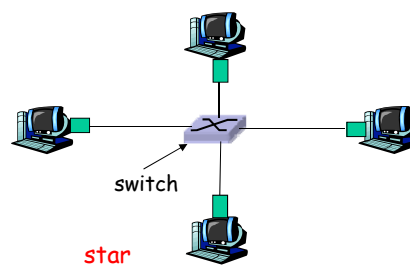
Star topology

- ❑ bus topology popular through mid 90s, and later star topology with **hub** at center
 - all nodes in same collision domain (their transmissions can collide with each other)
- ❑ today: star topology with **active switch** (layer 2) at center
 - no collision



bus: coaxial cable

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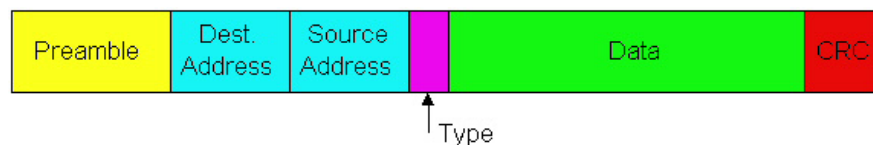


star

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Ethernet Frame Structure

Sending adapter encapsulates IP datagram (or other network layer protocol packet) in **Ethernet frame**



Preamble:

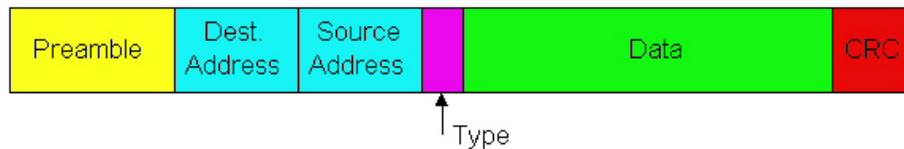
- ❑ 7 bytes with pattern 10101010 followed by one byte with pattern 10101011
- ❑ used to synchronize receiver, sender clocks
 - long preamble used due to "burst" nature of transmissions, unlike a synchronous point to point link

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Data Link Layer (SSL) 5-52

Ethernet Frame Structure (cont.)

- ❑ **Addresses:** 6 bytes
 - if adapter receives frame with **matching destination** address, or with **broadcast** address (eg ARP packet), it passes data in frame to network-layer protocol
 - **else** adapter discards frame
- ❑ **Type:** 2 bytes, indicates the higher layer protocol, ARP or IP (many others are supported such as Novell IPX and AppleTalk)
- ❑ **CRC:** 4 bytes, checked at receiver, if error is detected, the frame is simply dropped



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Data Link Layer (SSL) 5-53

Unreliable, connectionless service

- ❑ **Connectionless:** No handshaking between sending and receiving adapters
- ❑ **Unreliable:** receiving adapter doesn't send acks or nacks to sending adapter
 - stream of datagrams passed to network layer can have gaps
 - gaps will be filled only if app is using TCP
- ❑ Ethernet's MAC protocol: unslotted **CSMA/CD with binary backoff**
 - Interval for random retransmission doubles after every additional collision

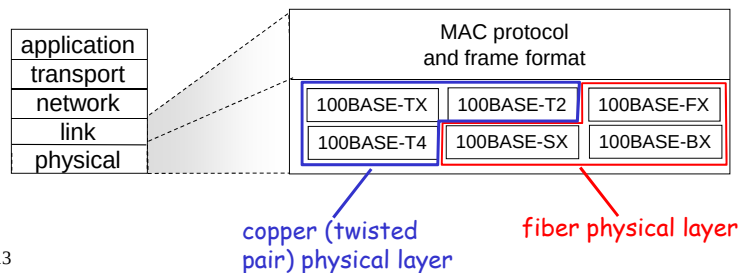
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Data Link Layer (SSL) 5-54

802.3 Ethernet Standards: Link & Physical Layers

❑ *many* different Ethernet standards

- same MAC protocol and frame format
- different speeds: 2 Mbps, 10 Mbps, 100 Mbps, 1Gbps, 10Gbps
- different physical layer media and technologies: cable, twisted pair, fiber



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Data Link Layer (SSL) 5-55

Link layer, LANs

5.1 introduction, services

5.2 error detection, correction

5.3 multiple access protocols

5.4 LANs

- addressing, ARP
- Ethernet
- *switches*
- *VLANs*

5.5 link virtualization: MPLS

5.6 data center networks

5.7 a day in the life of a web request
(play animation in .ppt slides on your own)

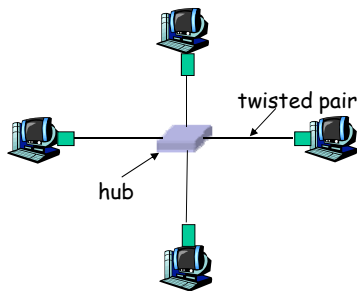
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Data Link Layer (SSL) 5-56

Hubs

Hubs are **physical-layer repeaters**:

- bits coming from one link go out all other links at the same rate
- no frame buffering -> **collisions like a coax cable**
- adapters detect collisions as usual



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Data Link Layer (SSL) 5-57

Switch (layer 2)

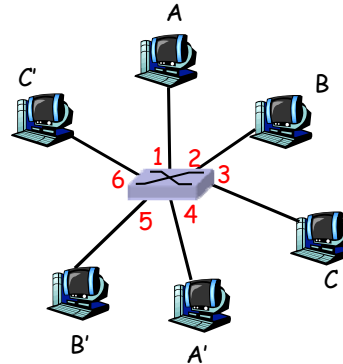
- Link layer device
 - **stores and forwards** Ethernet frames
 - examines frame header and **may selectively** forward frame to just one outgoing interface
 - **it still uses CSMA/CD** (just in case an outgoing interface is connected to a hub)
- **transparent**
 - hosts are unaware of presence of switches
- **plug-and-play, self-learning**
 - switches do not need to be configured

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Data Link Layer (SSL) 5-58

Switch: allows *multiple* simultaneous transmissions

- ❑ hosts have dedicated, direct connection (*full duplex*) to switch
- ❑ switch **buffers** packets
- ❑ **switching**: A-to-A' and B-to-B' simultaneously, without collisions
 - not possible with dumb hub
- ❑ Ethernet protocol used on *each* link



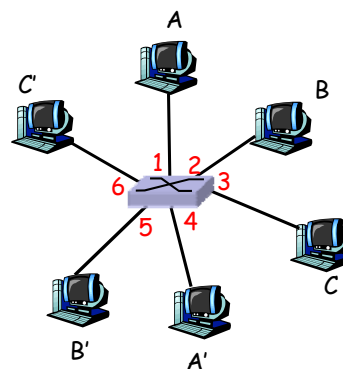
*switch with six interfaces
(1,2,3,4,5,6)*

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Data Link Layer (SSL) 5-59

Switch Table

- ❑ **Q**: how does switch know that A' reachable via interface 4, B' reachable via interface 5?
- ❑ **A**: each switch has a **switch table**, each entry:
 - (MAC address of host, interface to reach host, time stamp)
- ❑ looks like a forwarding table for routing
- ❑ **Q**: how are entries created, maintained in switch table?
 - no routing protocol is used



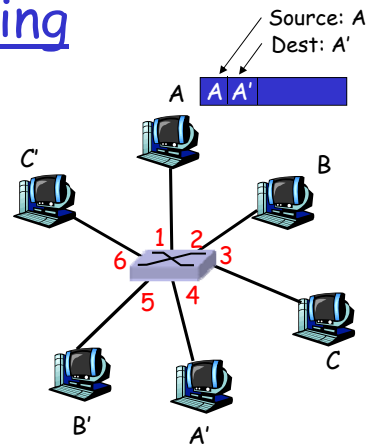
*switch with six interfaces
(1,2,3,4,5,6)*

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Data Link Layer (SSL) 5-60

Switch: self-learning

- switch *learns* which hosts can be reached through which interfaces
 - when frame received, switch "learns" location of sender: incoming LAN segment
 - records sender/location pair in switch table



What is required to make this work?

MAC addr	interface	TTL
A	1	60

Switch table (initially empty)

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Data Link Layer (SSL) 5-61

Switch: frame filtering/forwarding

When frame received:

1. record *interface* associated with sending host
2. index switch table using *MAC destination address*
3. **if** entry in table found for *destination*
 - then** {
 - if** dest is on interface from which frame arrived
 - then** drop the frame
 - else** forward the frame on interface indicated
 - }**
 - else** flood

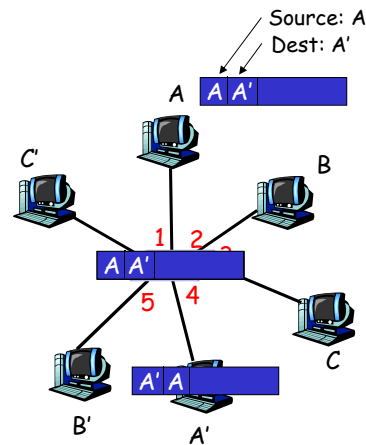
forward on all but the interface on which the frame arrived

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Data Link Layer (SSL) 5-62

Self-learning, forwarding: example

- destination A' unknown: *flood*
- destination A location known: *selective send*



MAC addr	interface	TTL
A	1	60
A'	4	60

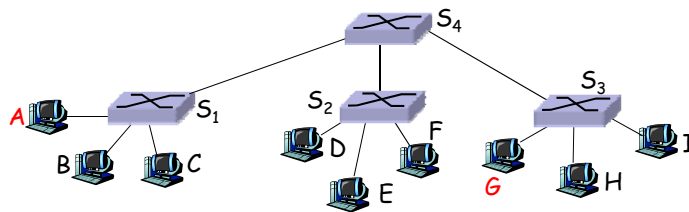
Switch table
(initially empty)

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Data Link Layer (SSL) 5-63

Interconnecting switches

- switches can be connected together



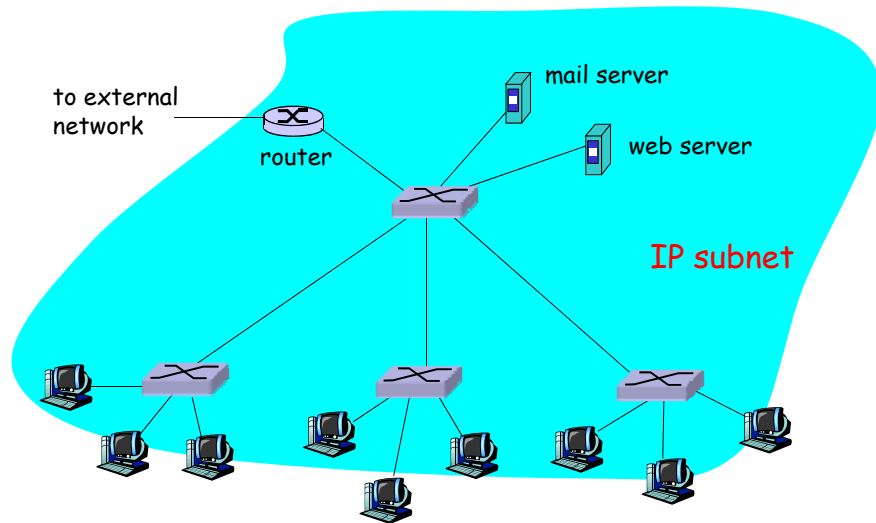
note: some links are idled if physical topology has loops

- Q: sending from A to G - how does S₁ know to forward frame destined to G via S₄ (and S₃) ?
- A: self learning (works exactly the same as in single-switch case)

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Data Link Layer (SSL) 5-64

Institutional network

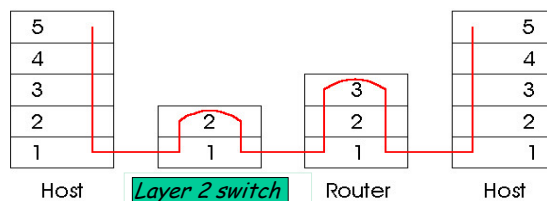


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Data Link Layer (SSL) 5-65

Switches vs. Routers

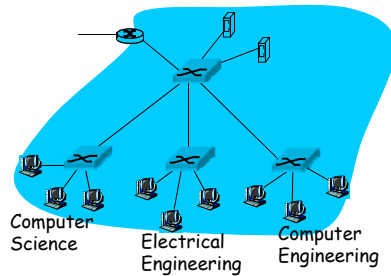
- both store-and-forward devices
 - routers: network layer devices (examine network layer headers)
 - Layer-2 switches are link layer devices
- routers maintain forwarding tables, implement routing protocols
- switches (layer 2) maintain switch tables, implement filtering, learning algorithms



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Data Link Layer (SSL) 5-66

VLANs: motivation



□ Suppose a single broadcast domain is configured

- all layer-2 broadcast frames (ARP, DHCP, etc.) cross entire LAN => security/privacy, efficiency issues

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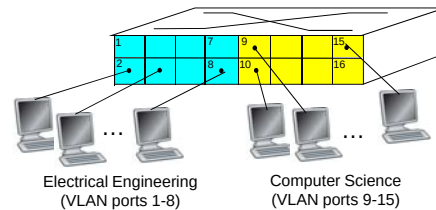
Data Link Layer (SSL) 5-67

VLANs

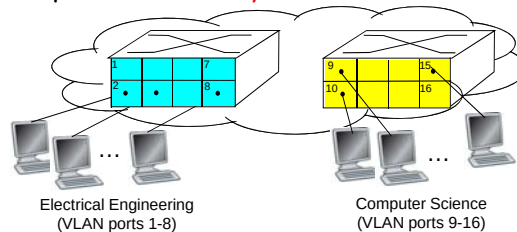
Virtual Local Area Network

A switch that supports VLAN capabilities can be configured to have multiple virtual LANs

Port-based VLAN: switch ports grouped (by switch management software) for a *single* physical switch which ...



... operates as *multiple* virtual switches

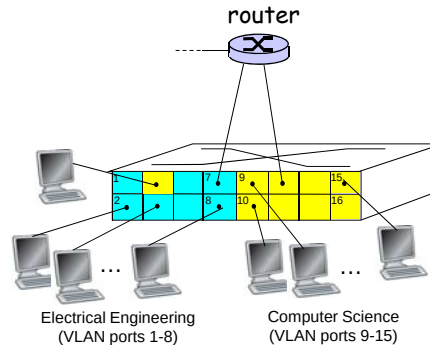


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Data Link Layer (SSL) 5-68

Port-based VLAN

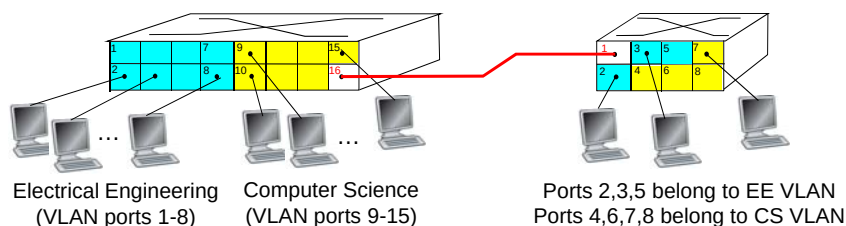
- ❑ **traffic isolation:** frames to/from ports of a VLAN can *only* reach its ports
 - can also define a VLAN based on MAC addresses of endpoints, rather than switch ports
- ❑ **dynamic membership:** ports can be dynamically assigned among VLANs
- ❑ **forwarding between VLANs:** done via a router (just as with separate switches)
 - ❖ in practice the router is built into the switch



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Data Link Layer (SSL) 5-69

VLANs spanning multiple switches



- ❑ **trunk ports:** carry frames between VLANs defined over multiple physical switches
 - frames forwarded within a VLAN between physical switches must carry VLAN ID info
 - 802.1q protocol adds/removes an additional header field for each frame forwarded between trunk ports

another example of tunnelling!

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Data Link Layer (SSL) 5-70

Link layer, LANs

5.1 introduction, services

5.2 error detection, correction

5.3 multiple access protocols

5.4 LANs

- addressing, ARP
- Ethernet
- switches
- VLANs

5.5 link virtualization: MPLS

5.6 data center networks

5.7 a day in the life of a web request

(play animation in .ppt slides on your own)

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Data Link Layer (SSL) 5-71

Link Virtualization: A Network as a Link

Virtual circuits provided by

- ATM, frame relay, which are packet-switching networks in their own right
 - with service models, addressing, routing different from Internet
- A subnet of MPLS capable routers

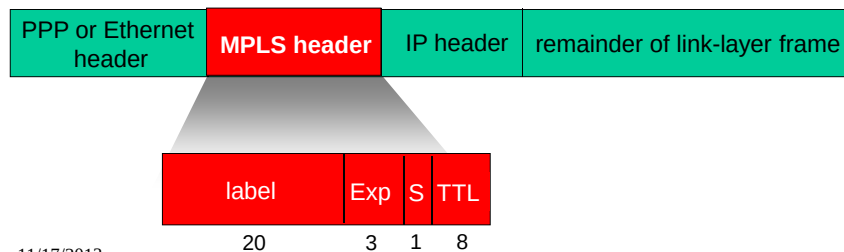
Each is "invisible" to IP and viewed as a link connecting two IP nodes

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Data Link Layer (SSL) 5-72

Multiprotocol label switching (MPLS)

- ❑ *initial goal*: speed up IP forwarding by using fixed-length label (instead of variable-length IP prefix) to do forwarding
 - borrowing ideas from Virtual Circuit (VC) approach
 - MPLS routers insert and remove MPLS header
 - but IP datagram still keeps IP address



Data Link Layer (SSL) 5-73

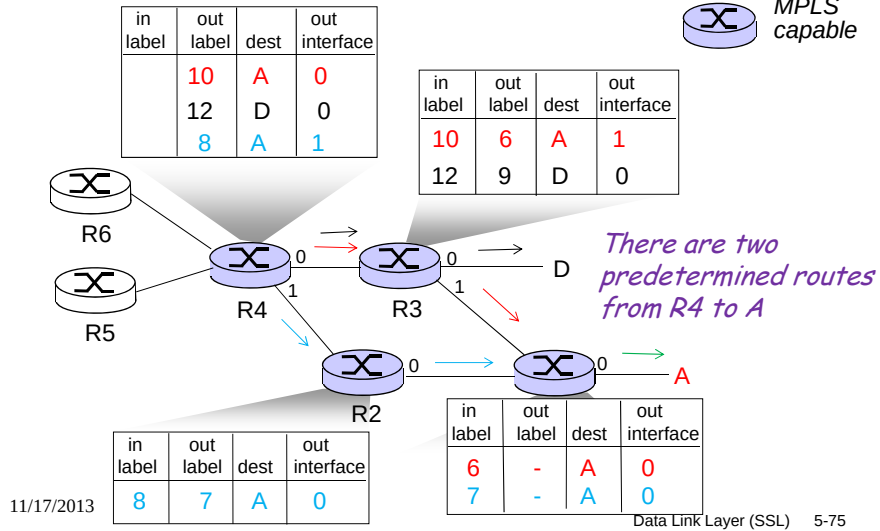
MPLS capable routers

- ❑ a.k.a. label-switched router
- ❑ forward packets to outgoing interface based only on **label value** (*does not inspect IP address*)
 - Much faster than longest prefix match
 - MPLS forwarding table distinct from IP forwarding tables
- ❑ **flexibility**: MPLS forwarding decisions can *differ* from those of IP

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Data Link Layer (SSL) 5-74

MPLS forwarding tables



MPLS applications

- ❑ **Fast failure recovery** - rerouting flows quickly to pre-computed backup paths (useful for VoIP)
- ❑ **Traffic engineering** - network operator can override IP routing and allocate traffic toward the same destination to multiple paths
- ❑ Resource provision for **virtual private networks**

Link layer, LANs

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(play animation in .ppt slides on your own)

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Data Link Layer (SSL) 5-77

Data center networks

□ 10's to 100's of thousands of hosts in close proximity supporting *cloud* applications

- e-business (e.g. Amazon)
- content-servers (e.g., YouTube, Akamai, Apple, Microsoft)
- search engines, data mining (e.g., Google)

❖ challenges:

- multiple applications, each serving massive numbers of clients
- managing/balancing load, avoiding bottlenecks in processing and networking



Inside a 40-ft Microsoft container, Chicago data center

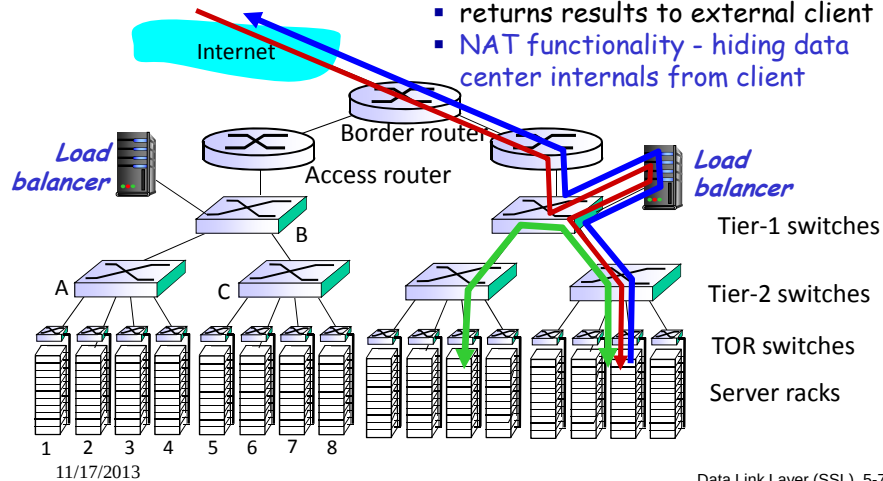
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Data Link Layer (SSL) 5-78

Data center networks

Load balancer:

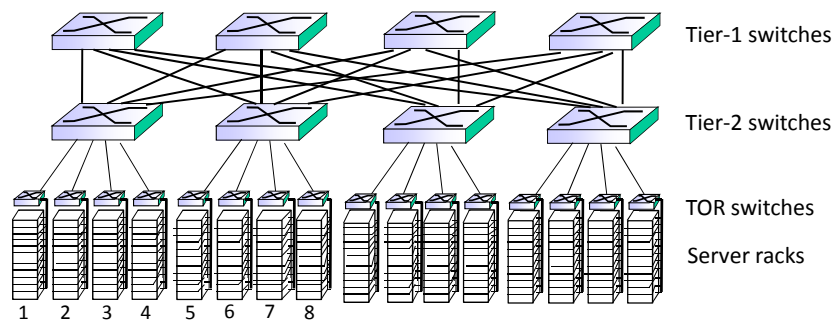
- receives external client requests for service
- directs workload within data center
- returns results to external client
- NAT functionality - hiding data center internals from client



Subnet below an access router

- rich interconnection among switches as well as duplication of routers/switches:

- increased reliability via redundancy
- increased throughput between racks (multiple routing paths possible)
- partitioned into smaller VLANs to localize ARP broadcast



focus of much recent research: routing, congestion control, etc.

Chapter 5: Summary

- ❖ principles behind data link layer services:
 - error detection, correction
 - sharing a broadcast channel: multiple access
 - link layer addressing
- ❖ instantiation and implementation of various link layer technologies
 - Ethernet
 - switched LANS, VLANs
 - virtualized networks as a link layer: MPLS
 - data center networks
- ❖ synthesis: a day in the life of a web request
(be sure to open Chapter5_A_Day_animation.ppt file on your own and see the animation)

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Data Link Layer (SSL) 5-81

The end

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Data Link Layer (SSL) 5-82